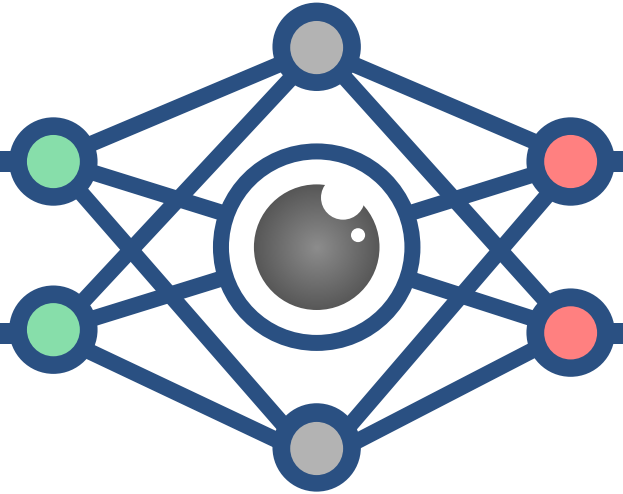
CS3485

Deep Learning for Computer Vision



Lec 12: Intro to MLOps

Announcements

- We'll have the quiz at the beginning of the lecture today!

(Tentative) Lecture Roadmap

Basics of Deep Learning

- Intro to Computer Vision
- Linear Classifiers and Perceptron
- Multilayer Perceptron
- Optimization in Deep Learning
- Pytorch I – MLPs
- Pytorch II – Images and Regularization
- Convolutional Neural Networks
- Data Augmentation and Deep CNNs
- Transfer Learning and Residual Nets
- Inception Net and what CNNs learn
- Adversarial Examples and Self-supervision

Deep Learning and Computer Vision in Practice

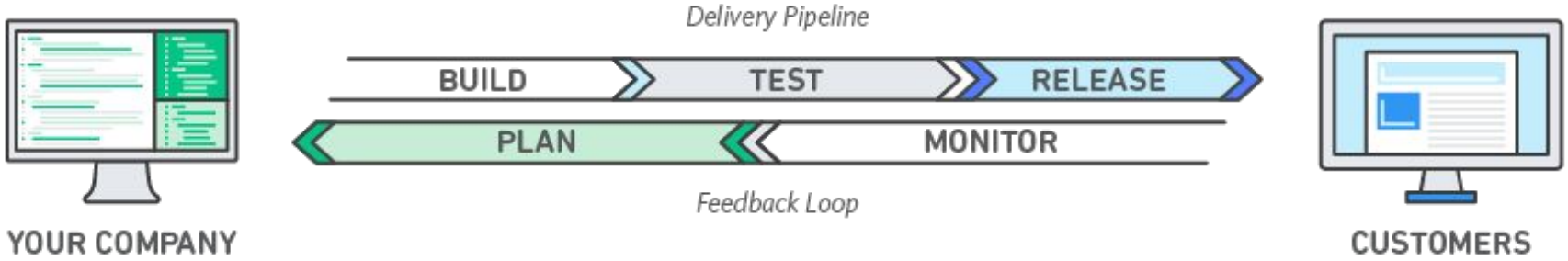
- Intro to MLOps
- Intro to Object Detection
- Fast Object Detection
- Intro to Image Segmentation
- Applications of Detection and Segmentation
- Autoencoders
- Image Generation with GANs
- Advanced GANs
- The Attention Mechanism
- Transformers and ChatGPT
- Image Generation by Prompt
- Misc. Topics

Deep Learning Theory vs Practice

- So far, we've seen some of the theory of Deep Learning and applied it to the problem of Image Classification.
- But obviously, most of Deep Learning's societal impact comes from its usage in the industry.
- Today, we'll dive a little bit on how these models are used in practice and how they are inserted in the software production line
- In order to know that, we'll dive into the world of **MLOps**.

A little digression: DevOps

- In any software company, there are two teams: **development** and **operations**:
  - The Development team plans, builds, tests new systems,
  - The Operations team implements and releases the products, and monitors user experience.
- These teams then create a **delivery pipeline** and a **feedback loop** specific to each team.
- However, often these teams may step on each other toes:
  - Sometimes, the operations team may provide feedback on bugs that need to be immediately fixed, causing delays in the development cycle.
  - Sometimes, the development “gets stuck” due to operational slowness.

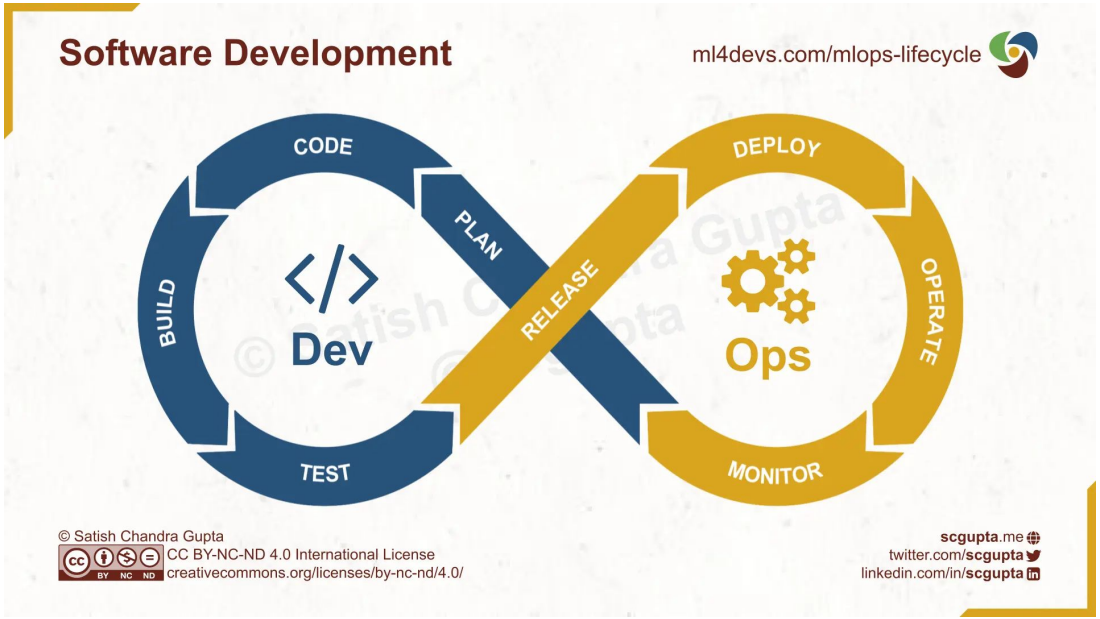


A little digression: DevOps

- **DevOps** unites Development and Operations teams, breaking down traditional silos between them and serves multiple purposes:
  - Faster launch of new features.
  - Increased customer and developer satisfaction through efficient processes,
  - Feedback loops for better communication.
- DevOps shifts the focus from team to **organizational goals**, from finger-pointing to **collective ownership**, enabling development and operations teams to work together seamlessly.
- It also enables the implementation of **CI/CD** processes:
  - **Continuous Integration (CI):** Developers frequently commit code changes that are automatically built, packaged, and tested through an automated pipeline.
  - **Continuous Delivery/Deployment (CD):** Code changes that have passed through the CI pipeline are automatically prepared for deployment and can be rapidly and safely released to production environments.

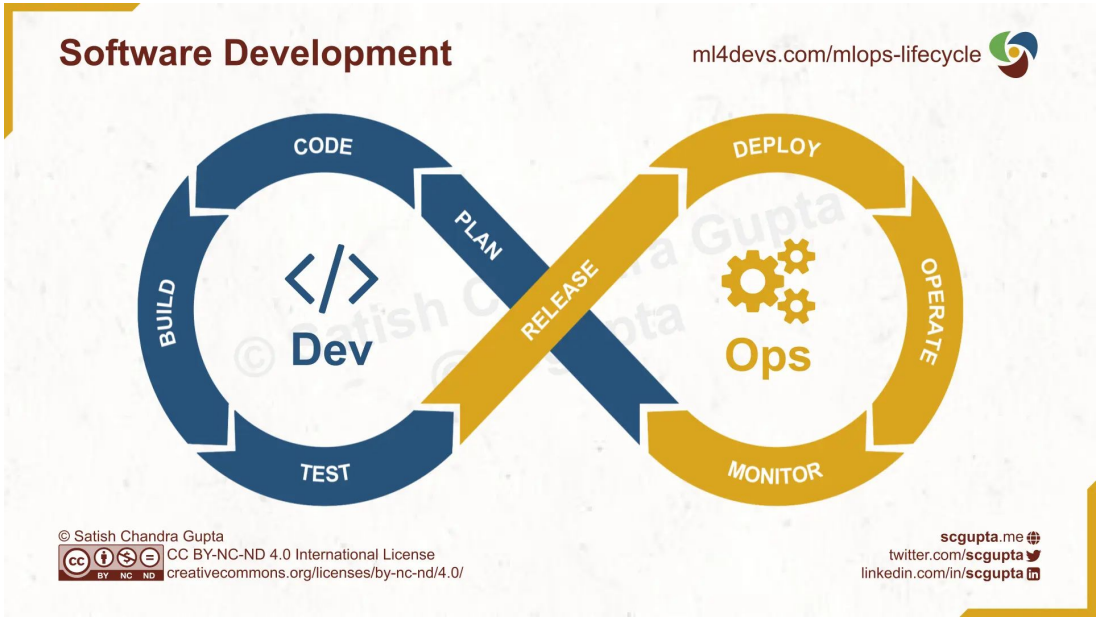
A little digression: DevOps LifeCycle

- In practice, companies implement the famous **DevOps lifecycle,** which consists of eight main stages represented as an infinity loop:
  - **Planning**: You collect the end-user data and create a roadmap of future processes at this stage.
  - **Coding**: At this stage, developers use tools or plugins to streamline the development process.
  - **Building**: Once developers finish coding, they commit the code to the shared repository.



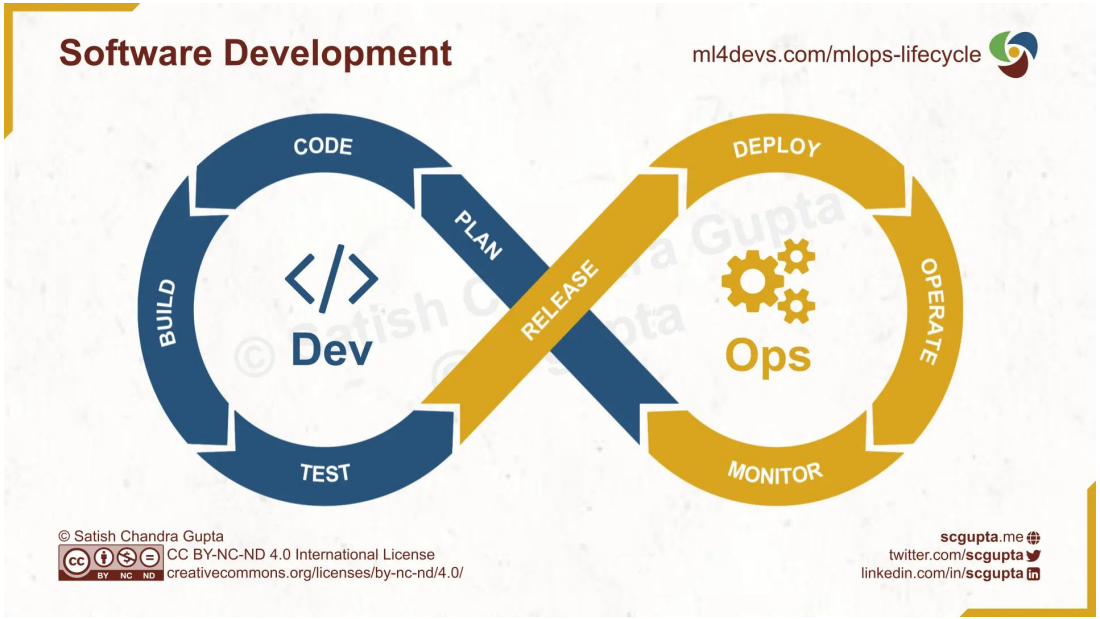
A little digression: DevOps LifeCycle

- In practice, companies implement the famous **DevOps lifecycle,** which consists of eight main stages represented as an infinity loop:
  - **Testing**: Look for bugs at all levels, i.e, unit tests, integration tests, coverage tests, performance tests, load tests, privacy tests, security tests, and bias tests.
  - **Releasing**: The DevOps team makes sure the build has passed all necessary tests in accordance with organisational needs.



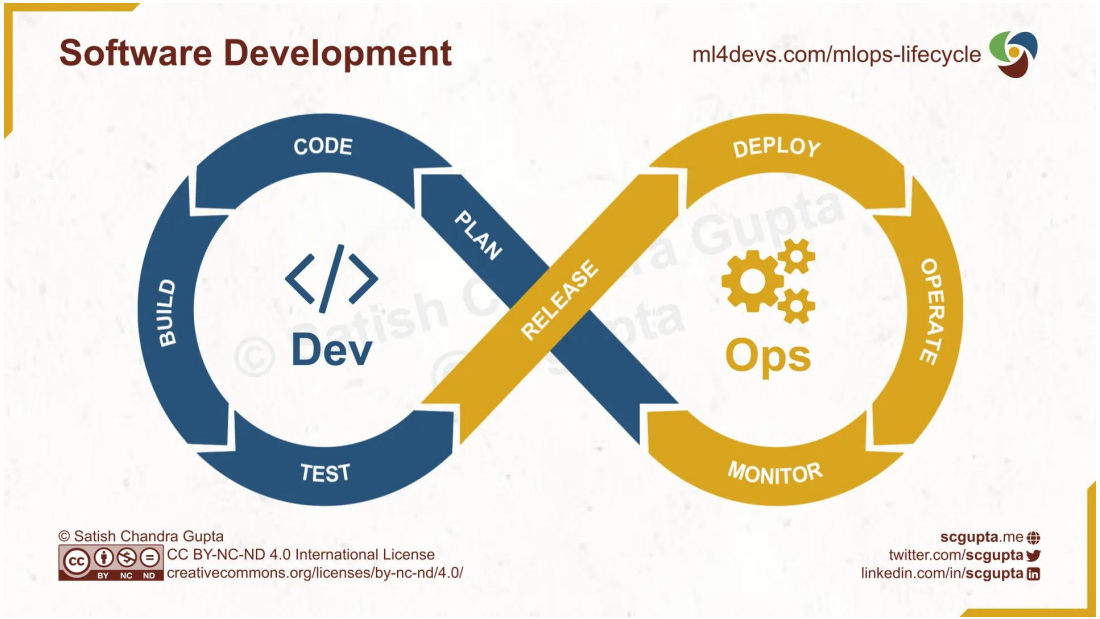
A little digression: DevOps LifeCycle

- In practice, companies implement the famous **DevOps lifecycle,** which consists of eight main stages represented as an infinity loop:
  - **Deployment**. You need to create the production environment and release the build at this stage. When deploying, we usually *create an endpoint* to serve our product.
  - **Operating**. Here, your product is ready for use by clients, and the operation team is still managing server configurations.



A little digression: DevOps LifeCycle

- In practice, companies implement the famous **DevOps lifecycle,** which consists of eight main stages represented as an infinity loop:
  - **Monitoring**. You can collect the data from customer experience and the first use of the product and use it for improving productivity and fixing possible bugs.
- This process then repeats and cycles continuously, as bugs are fixed, and company priorities change, without impacting product quality and processes.



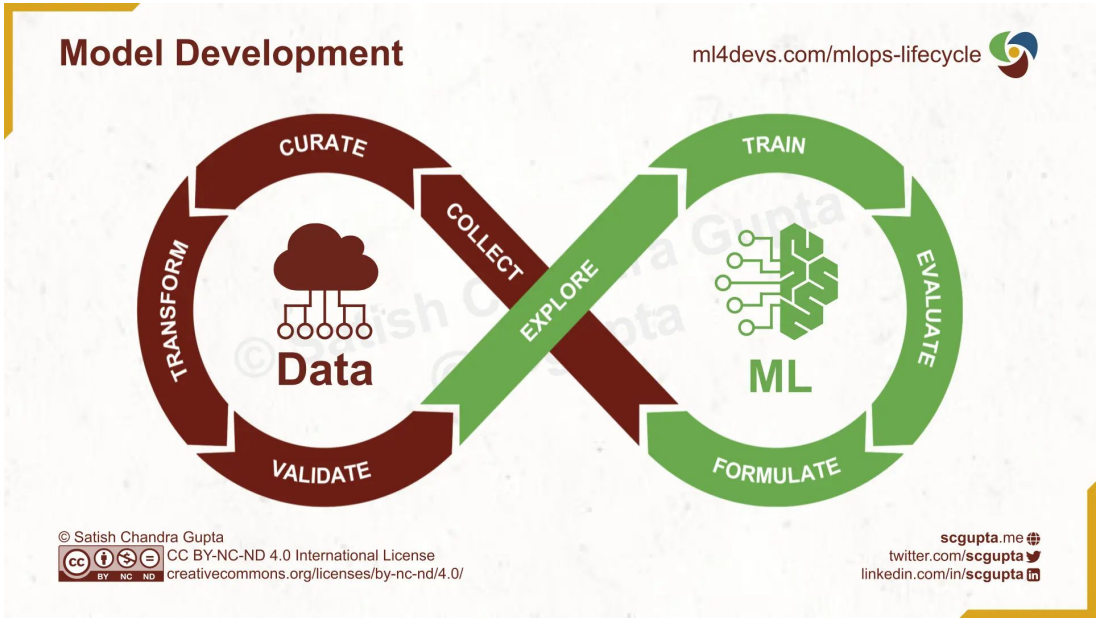
MLOps

- With the rise in importance of Machine Learning (and Deep Learning in particular) in the software industry, typical ML pipelines started being incorporated in DevOps.
- Indeed, ML problems are found in many industrial sectors. For example:
  - In **finance**, software may to handle fraud detection, risk management and algorithmic trading.
  - In **manufacturing**, there is supply chain optimization, demand forecasting and quality control.
  - In **media and entertainment**, we find content recommendation and sentiment analysis.
- That gave rise to what is called **MLOps** to streamline and enhance the deployment, monitoring, and management of machine learning models
- The field of MLOps is not yet as established as DevOps, but many companies are already **eager to hire** professionals in that ML/Dev/Ops intersection.

A typical MLOps engineer salary according to Glassdoor.

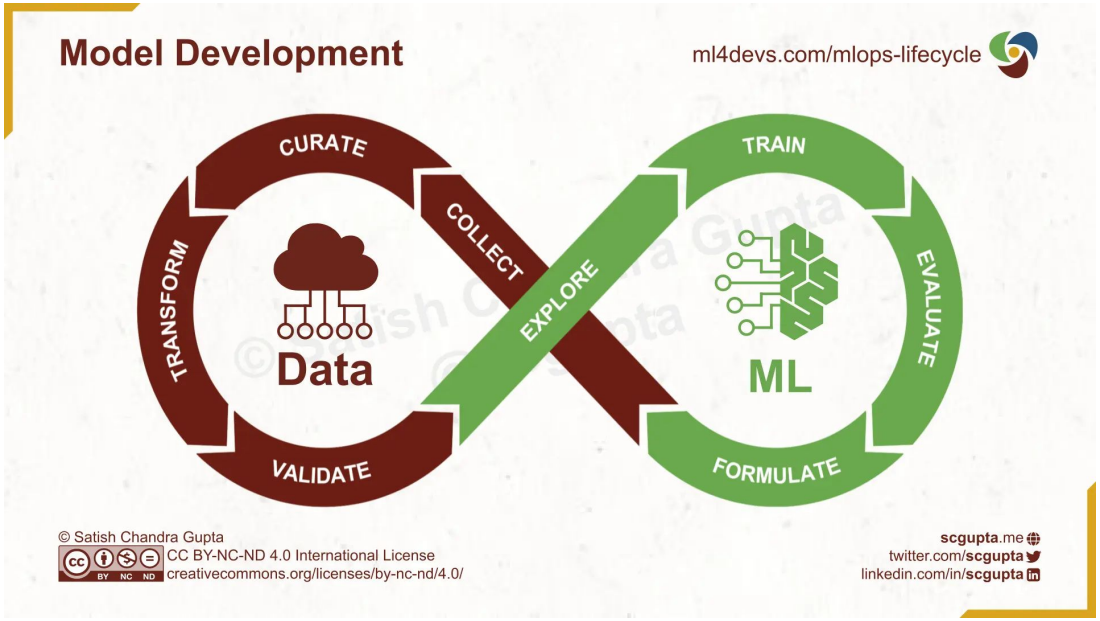
MLOps Lifecycle

- Just like with DevOps, MLOps also has a lifecycle that consists of many data driven stages. One example of such has the following stages:
  - **Formulate:** Translating a business objective into a machine learning problem. Some factors to consider:
    a. *Cost of mistakes*: How costly are model mistakes?
    b. *Data*: is the data available/cheap?
    c. *Evaluation Metrics*: What metric do you want to optimize?



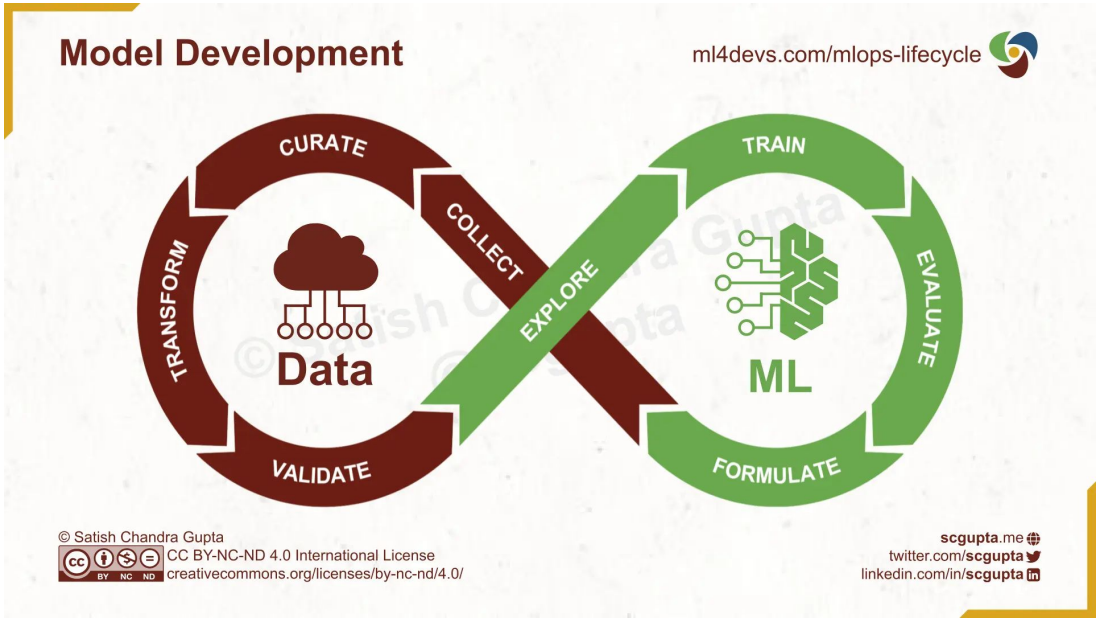
MLOps Lifecycle

- Just like with DevOps, MLOps also has a lifecycle that consists of many data driven stages. One example of such has the following stages:
  - **Collect:** collect the necessary data from internal applications as well as external sources.
  - **Curate**: Collected data is almost never pristine. You need clean it, remove duplicates, fill in missing values, and store it.
  - **Transform**: You transform the cleaned data to suit your ML modeling.



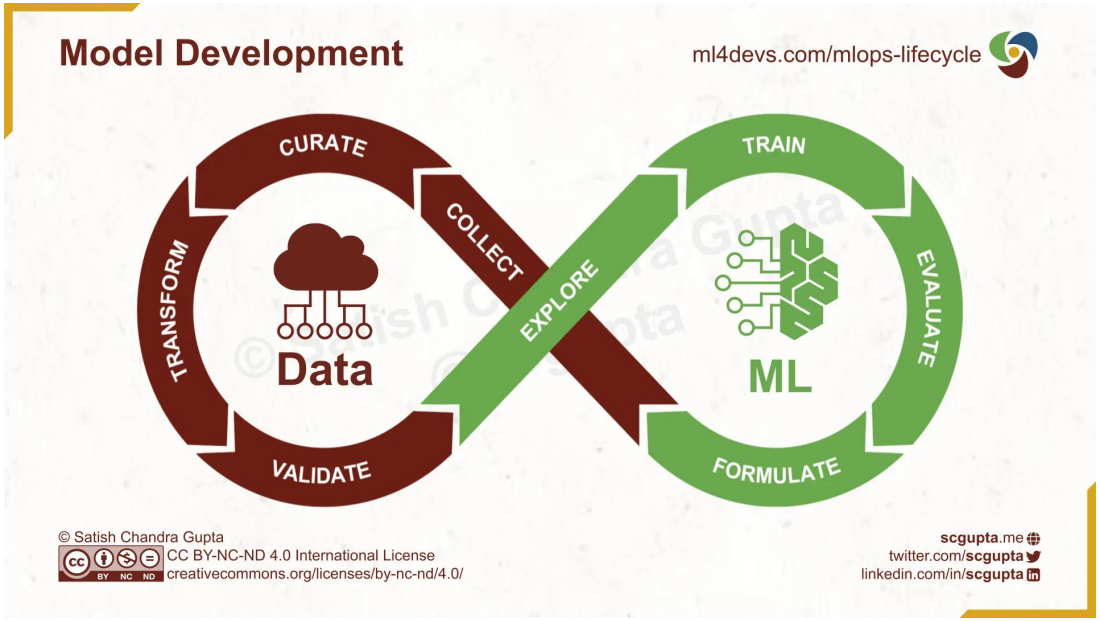
MLOps Lifecycle

- Just like with DevOps, MLOps also has a lifecycle that consists of many data driven stages. One example of such has the following stages:
  - **Validate:** Implement quality checks, maintain logs of data distributions over time, and create triggers to alert when any of the checks fail.
  - **Explore**: ML engineers perform data analysis to understand the relationships between various features and the target values.



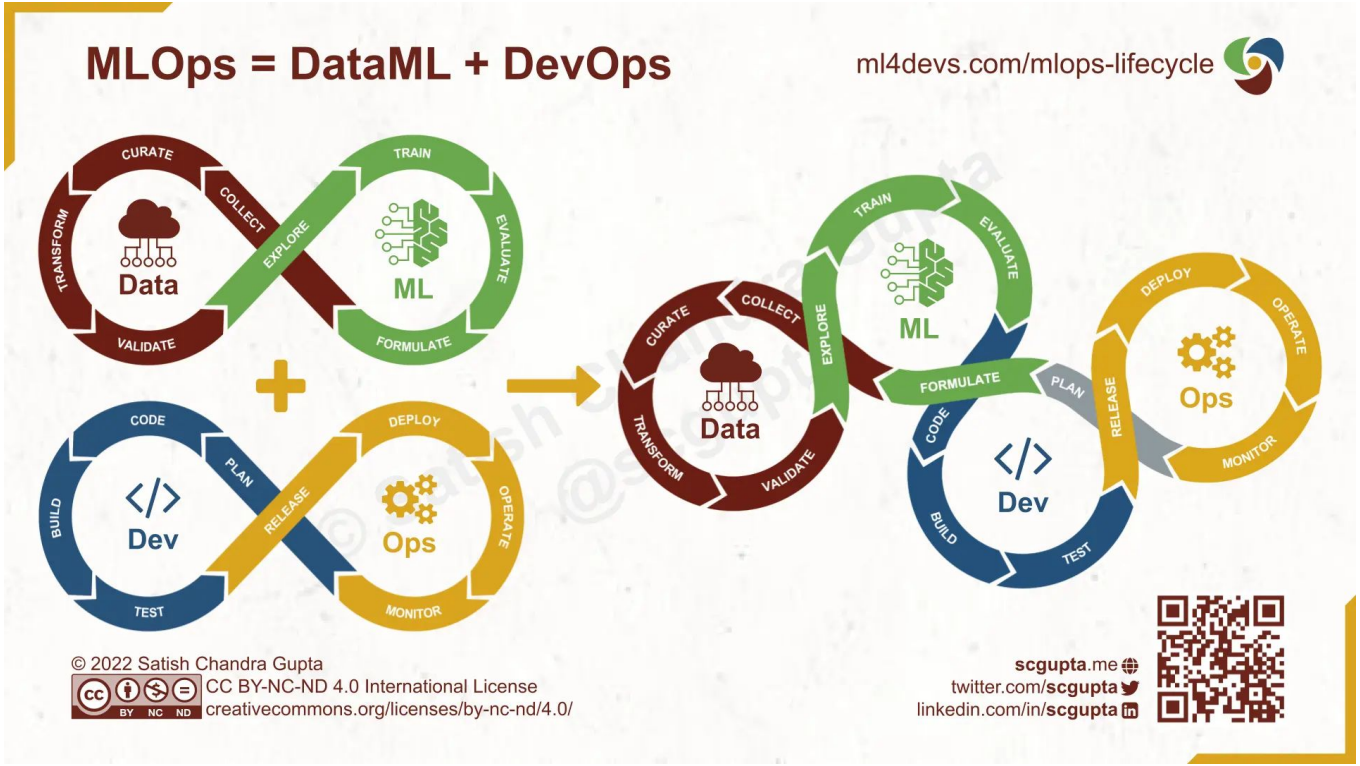
MLOps Lifecycle

- Just like with DevOps, MLOps also has a lifecycle that consists of many data driven stages. One example of such has the following stages:
  - **Train:** ML engineers train multiple models, run experiments, compare performances, tune hyper-parameters, and select a couple of best-performing models.
  - **Evaluate**: Evaluate the model characteristics against business objectives and pre-established metrics.



DevOps and MLOps

- In practice, ML model and software development are becoming more and more tied together into one larger eternal **MLOps knot** (which some define as the true MLops lifecycle).



Example of deployment of an ML model

- The connection between model training and deployment is crucial, and we won't have a lot of time in this course to see how this happens in practice.
- However, I can at least show you a bit what a simple deployment of one of our previous codes looks like.
- Here, we'll use the **Cat vs. Dog classification problem**, solved using transfer learning on a small network architecture called MobileNet implemented in Pytorch.
- For deployment, we'll use a Python library called **Flask**, a web framework that allows developers to build simple (and also complex) web applications very easily.
- Let's see how it works in practice (and here's the code if you want to try it yourself)

Pytorch Code
(for training the model)

Click here to open code in Colab

Flask Code
(for deploying the model)

Click here to open code in Github

Example of deployment of an ML model

Cat vs. Dog Classifier

Upload Image | Predict the Image

Classification Results

This is a cat (99.58% confidence)!

Cat vs. Dog Classifier

Upload Image | Predict the Image

Classification Results

This is a dog (99.2% confidence)!

Cat vs. Dog Classifier

Upload Image | Predict the Image

Classification Results

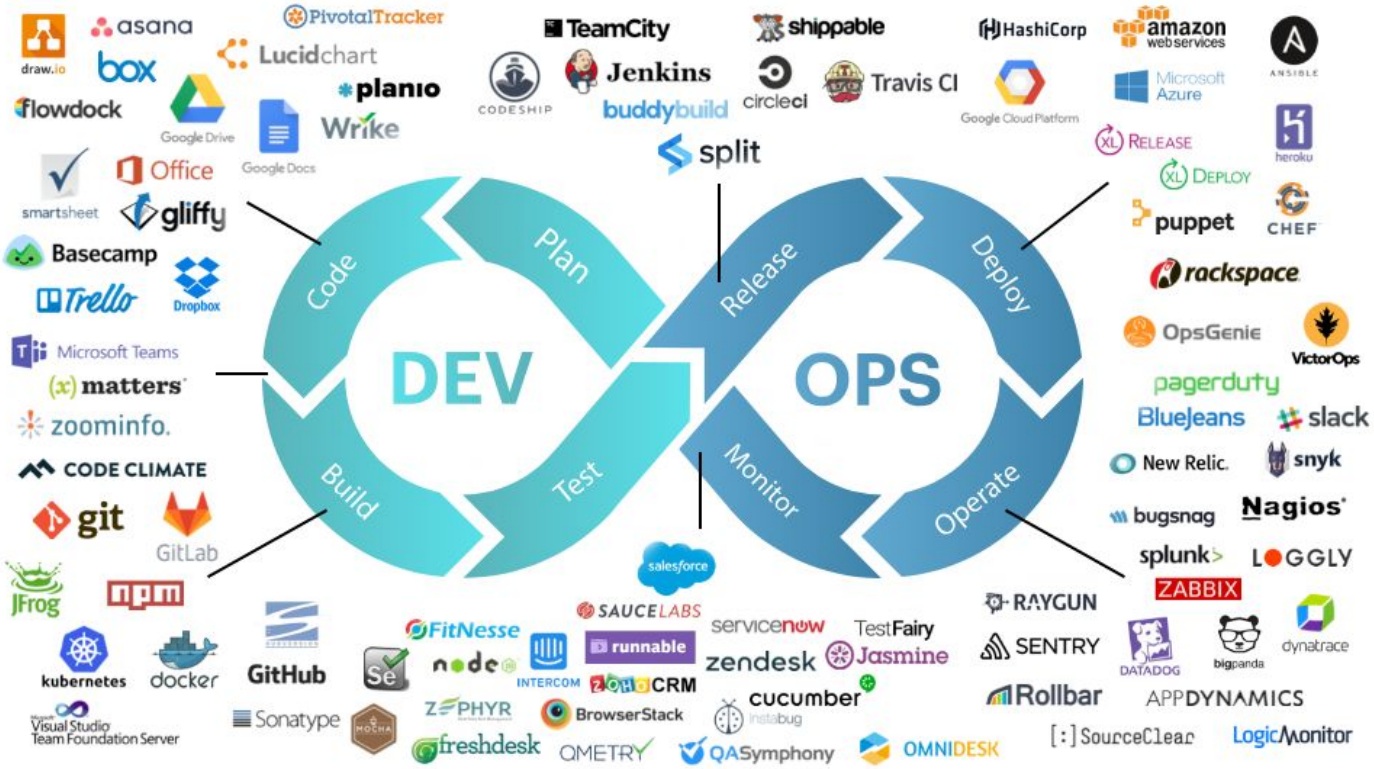
This is a other (74.35% confidence)!

Exercise (in pairs)

- Think of an industry and a particular task within that industry may take advantage of using ML in its processes. Then design an MLOps pipeline (the *formulate* step, in particular, but you can brush the others as well) that continuously improve and maintains that pipeline. *For example*: say I want to predict how good my lectures are using ML and use it to improve future classes. Then:
  - *Formulate*: ML mistakes are not so costly, I can use my slides or recordings as input data and students evaluations as output data, the metric could be whether I reached 70% of overall satisfaction.
  - *Train*: Each time, at the end of the semester, my software automatically loads my new slides and recordings and retrains the prediction network.
  - *Evaluate*: I can then test my newly trained model on data from years where I know the class was successful and those where it wasn't.

Tools for DevOps

- In DevOps, it's common to have (many) specific tools for each stage of its lifecycle.
- Mastering at least few of them is essential to the modern developer.



A case study: Vertex AI

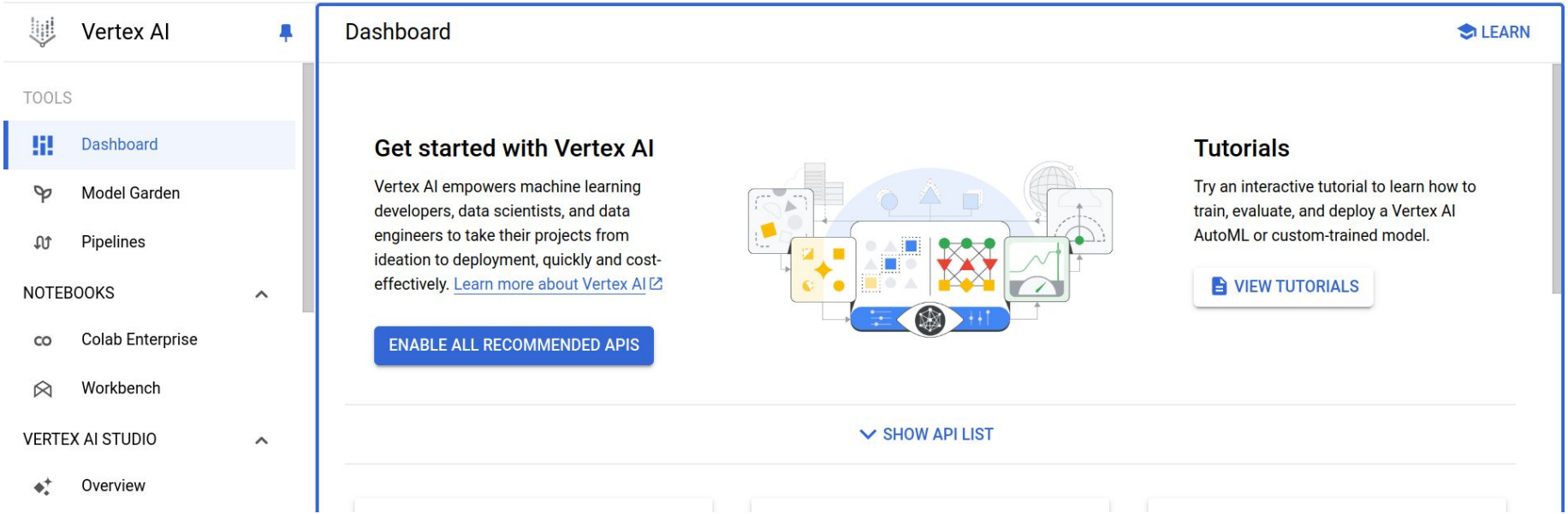
- There are also specific tools that can be used solely when automatizing MLOps processes.
- Nowadays, the most useful ones are embedded in Machine Learning platforms with the biggest cloud computing providers*:
  - Microsoft's Azure provides **Azure ML**
  - Amazon's AWS (Amazon Web Services) provides **SageMaker**
  - Google's GCP (Google Cloud Platform) provides **Vertex AI**.
- Vertex AI as its platform is in some ways similar to Colab's**. Let's see what it provides.

Google's GCP Vertex AI

* Although some smaller companies are also starting to take a larger share of that maket.
** Here's a good summary of Vertex AI's tools for MLOps.

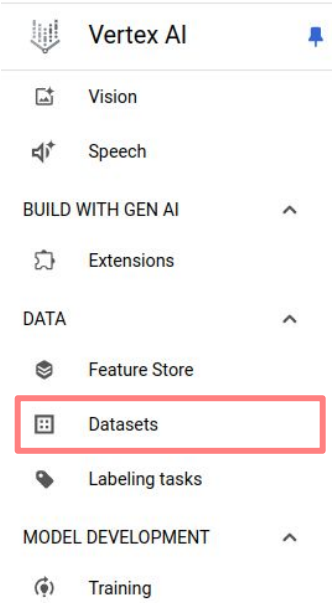
A case study: Vertex AI

- After logging in GCP, we can hit “console” for some project and search for Vertex AI. We’ll then see the following dashboard:



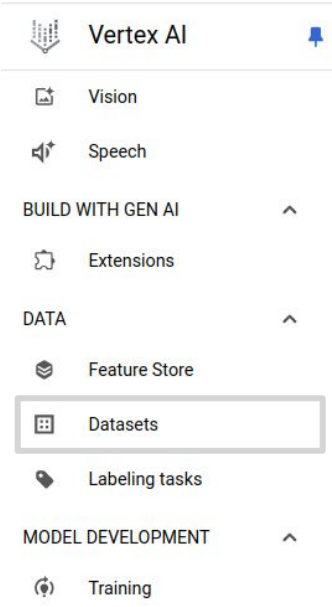
A case study: Vertex AI

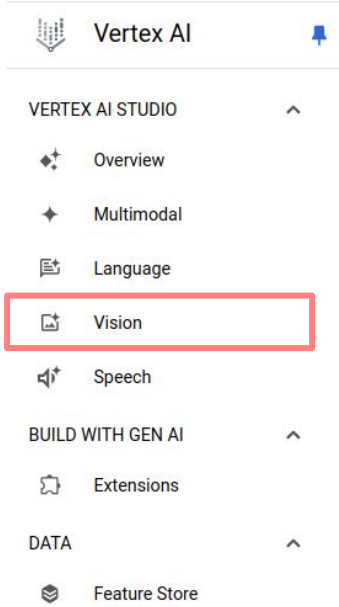
- At Vertex AI, you have tools to: **manage datasets**



A case study: Vertex AI

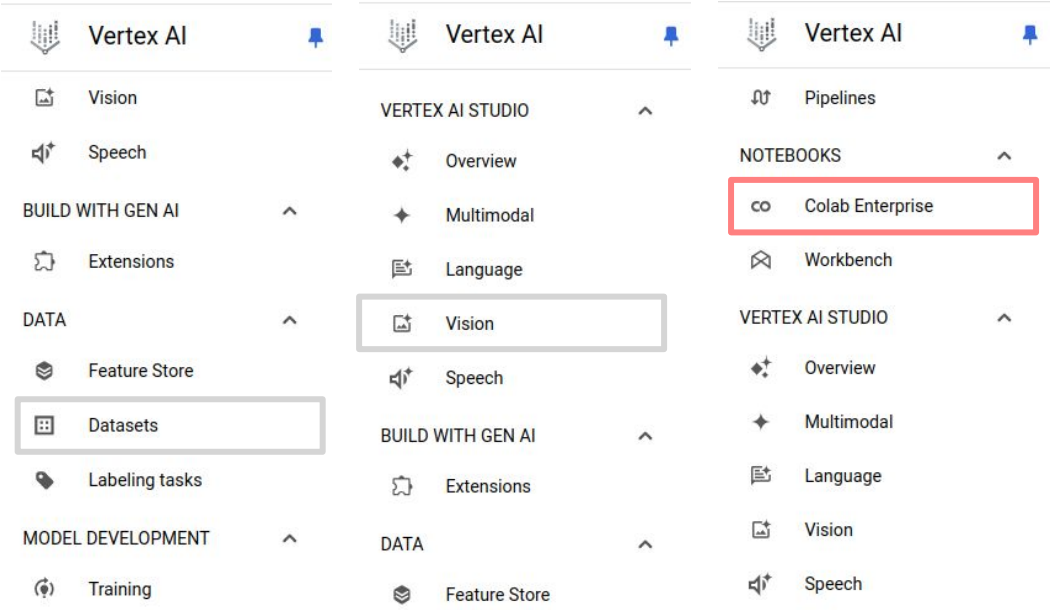
- At Vertex AI, you have tools to: manage datasets, **use pretrained vision (or language) models**





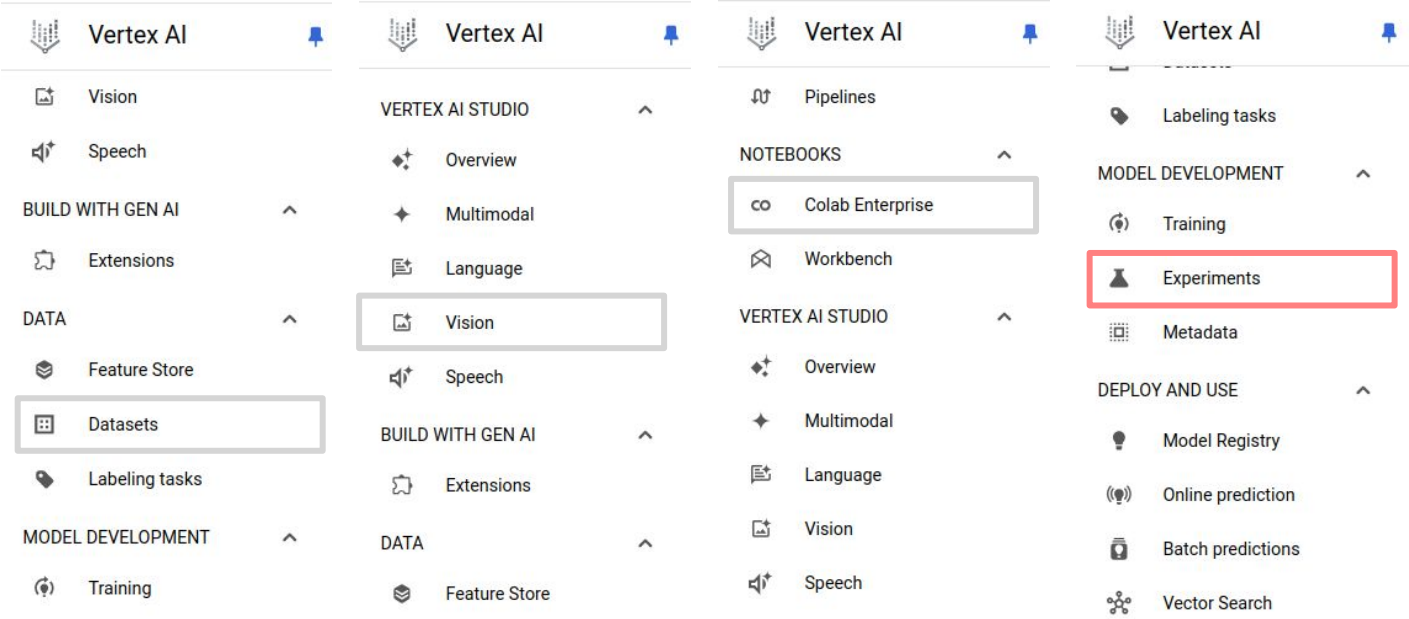
A case study: Vertex AI

- At Vertex AI, you have tools to: manage datasets, use pretrained vision (or language) models, **train your own models**



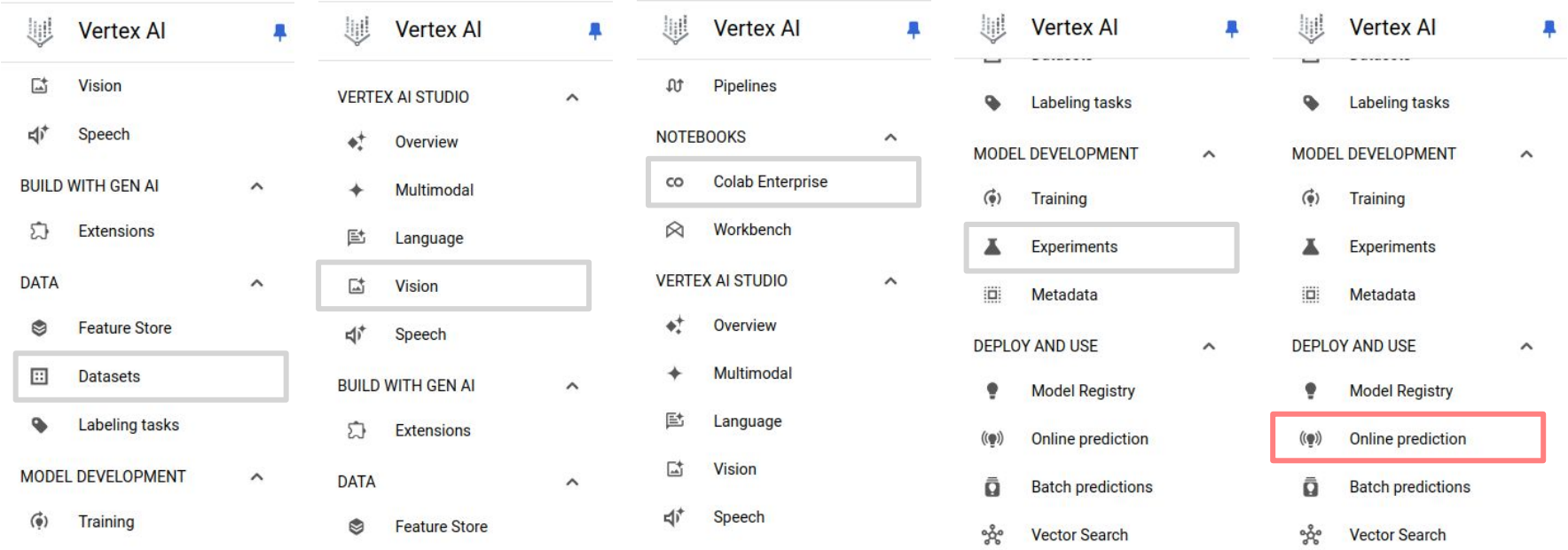
A case study: Vertex AI

- At Vertex AI, you have tools to: manage datasets, use pretrained vision (or language) models, train your own models, **design experiments**



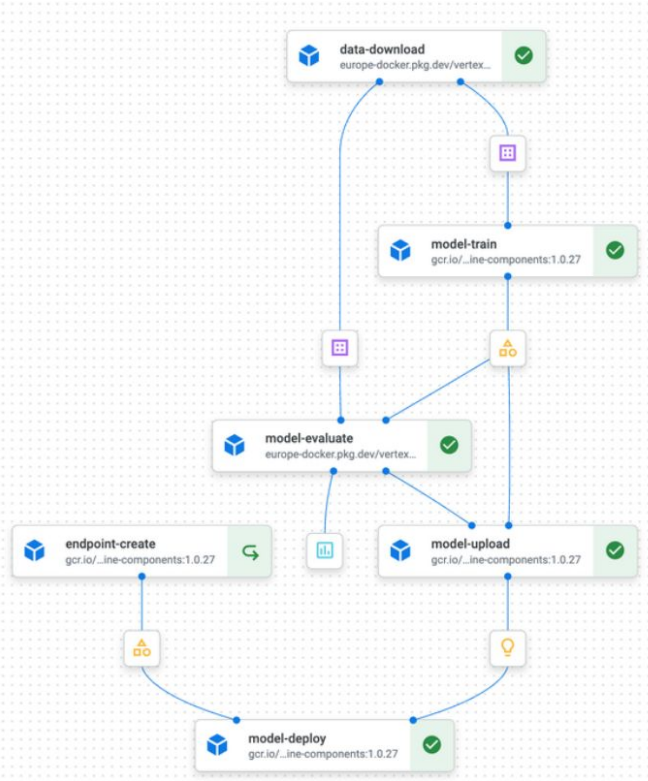
A case study: Vertex AI

- At Vertex AI, you have tools to: manage datasets, use pretrained vision (or language) models, train your own models, design experiments, **deploy models to endpoints**, etc.



A case study: Vertex AI

- At the industrial level, developers **orchestrate** the whole MLOps life cycle in Vertex AI **programmatically** (i.e., via code, not the dashboard) using what are called Pipelines*.
- Here you have a pipeline created with Vertex AI's SDK (software development kit), which automatically:
  - Downloads data and trains the model with the data.
  - Evaluates and stores the model.
  - Deploys it to an API endpoint.
- Once you have the endpoint, you can connect it with any programmable services, from mobile and web apps to operating systems.



* Here's a great video from this great channel called MikeStat that covers the whole Pipeline Orchestration in Vertex AI. The video also comes with a great github repo.

Final Project Specifics

- One of your final project types will have an MLOps theme:
  - Pick a problem in an industry of your interest that can use Deep Learning in its processes.
  - Plan out how to implement each step of MLOps life cycle for that problem.
  - Implement some or all of those steps in practice.
  - Deploy the resulting model.
- A few notes here:
  - You'll can use any pretrained model you find online or in our slides (the focus won't be on the model *per se*).
  - You'll be mostly graded on the quality of your product.
  - Take this as something to put on your resume!

(Machine) Learning in an Ever Changing World

- If you choose that theme for your final project, you'll have to learn many new tools and concepts by yourself.
- This is somewhat intentional as well: **learning how to learn*** is essential to *any* intellectual endeavor and it is particularly crucial in modern software development.

* The following advices are from Yonas Gebregziabher, a Bowdoin CS alumni who's currently working at Microsoft as a developer.

(Machine) Learning in an Ever Changing World

- There are a few reasons for why that is true in the industry:
  - *Technology evolves quickly*:
    - Frameworks get updates, new technologies are developed, and the industry often quickly responds to these changes.
    - You need to also **pick up on these changes** and be ready to implement them, should they significantly impact your product.
  - *Re-orgs are very common* (*especially in larger companies*):
    - Changes in management and shifts in business needs cause entire teams to abandon projects and reshuffle. This means you might find yourself on a new team with a tech stack you're not entirely familiar with.
    - You need to quickly **adapt to new teams** so that you can begin to contribute as well!
  - *The majority of a developer time is spent reading code not writing it*:
    - Oftentimes, you're maintaining an existing codebase, adding features, and fixing bugs.
    - You need to be able to develop skills to **comprehend code bases** so that you can quickly ramp up. Here is a good read on it.

How to learn how to learn?

- For these reasons, you need to be able to read code and read documentation often to learn new things. **Learning is at the core of a software developer's career**.
- But how do you learn? Here a few tips:
  - You need to figure out what works best for you:
    - Oftentimes, the best approach to learning something new is to simply just build something that incorporates whatever you want to learn.
    - You learn best by struggling to figure out a bug while working on a project that interests you.
  - It is completely okay to get started with YouTube videos that introduce you to these concepts,
    - However, it is counterproductive if you're copy-pasting everything you see in the videos all the time.
    - You'll get stuck in **tutorial hell**, where it's hard for you to reuse the technology you learned outside of the context of the video you followed. Here is a good read on tutorial hell and how to escape it.

How to learn how to learn?

- By building projects you find interesting, you do a couple of things:
  a. You're more motivated to keep working on the project through the difficult stages. This keeps you wanting to learn more.
  b. You truly learn what you're using to build your project because you've understood it enough to apply it to your use case.
  c. You learn best by failing time and time again by working on stuff you find interesting.
- After years of this, picking up new things becomes easier because you've built a habit of learning that works best for you.

Video: AI Art Generation